

Es. 1) Quick sort

14 9 13 38 10 24 8 25 15 18 3 12 16 17

SVOLGIMENTO:

1ª Fase)

14	9	13	38	10	24	8	25	15	18	3	12	16	17
•												•	
14	9	13	38	10	24	8	25	15	18	3	12	16	17
			•									•	
14	9	13	16	10	24	8	25	15	18	3	12	38	17
					•						•		
14	9	13	16	10	12	8	25	15	18	3	24	38	17
						•				•			
14	9	13	16	10	12	3	25	15	18	8	24	38	17
							•						
14	9	13	16	10	12	3	15	25	18	8	24	38	17
							•	•					

1° Pivot
• = i
• = j

i e j si sono invertiti: scambio pivot con l'elem all'indice i e ricarro.

14 9 13 16 10 12 3 15 17 18 8 24 38 25

2ª Fase)

14	9	13	16	10	12	3	15	17	18	8	24	38	25
			•			•							
14	9	13	3	10	12	16	15	17	18	8	24	38	25
					•	•							
14	9	13	3	10	12	15	16	17	18	25	24	38	8
													•

i e j si sono invertiti

i e j si sono invertiti

Es. 2) Parentesiizzazione di una

$A_1 = 5 \times 9$, $A_2 = 9 \times 2$, $A_3 = 2 \times 7$, $A_4 = 7 \times 4$

m:

	A_1	A_2	A_3	A_4
A_1	0	90	160	126
A_2		0	126	128
A_3			0	56
A_4				0

S:

	A_1	A_2	A_3	A_4
A_1	0	1	2	2
A_2		0	2	2
A_3			0	3
A_4				0

$A_1 \times A_2 = 5 \cdot 9 \cdot 2 = 90$ $A_2 \times A_3 = 9 \cdot 2 \cdot 7 = 126$

$A_3 \times A_4 = 2 \cdot 7 \cdot 4 = 56$

$A_1 \times A_2 \times A_3$: $K=2 \rightarrow A_1 \times (A_2 \times A_3) = 0 + 126 + 5 \cdot 9 \cdot 7 = 441$

$K=2 \rightarrow (A_1 \times A_2) \times A_3 = 90 + 0 + 5 \cdot 2 \cdot 7 = 160$

$A_2 \times A_3 \times A_4$: $K=2 \rightarrow A_2 \times (A_3 \times A_4) = 0 + 56 + 9 \cdot 2 \cdot 4 = 128$

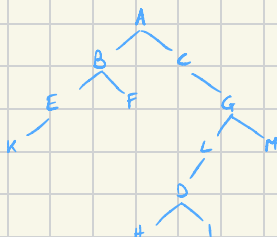
$K=3 \rightarrow (A_2 \times A_3) \times A_4 = 126 + 0 + 9 \cdot 7 \cdot 4 = 378$

$A_1 \times A_2 \times A_3 \times A_4$: $K=2 \rightarrow A_1 \times (A_2 \times A_3 \times A_4) = 0 + 128 + 5 \cdot 9 \cdot 4 = 308$

$K=2 \rightarrow (A_1 \times A_2) \times (A_3 \times A_4) = 90 + 56 + 5 \cdot 2 \cdot 4 = 186$

$K=3 \rightarrow (A_1 \times A_2 \times A_3) \times A_4 = 441 + 0 + 5 \cdot 7 \cdot 4 = 300$

Es. 3) Visitare in pre-order, post-order, in-order



SVOLGIMENTO:

- Pre-order: A, B, E, K, F, C, G, L, D, H, I, M

- In-order: K, E, B, F, A, C, H, D, I, L, G, M

- Post-order: K, E, F, B, H, I, D, L, M, G, C, A

Es. 4) Tabella hash, $N=7$. chiavi: 40, 50, 106, 62, 215, 30, 36. Quadratic probing, $c_1=1$, $c_2=1$
 $\alpha < 0.5$

SVOLGIMENTO:

$$\alpha = \frac{N}{M} \rightarrow M = \frac{N}{\alpha} = \frac{7}{0.5} = 14 \rightarrow M=17. \quad \text{quadratic probing: } h(K) = (K \cdot \alpha + c_1 \cdot i + c_2 \cdot i^2) \% M$$

0	96	7		14	
1		8		15	30
2		9		16	50
3		10			
4	140 (106 hit!)	11	62 (215 hit!)		
5		12			
6	106	13	215 (30 hit!)		

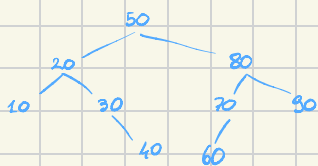
$$106 \% 17 = 4 \quad \text{hit} \Rightarrow (4 + 1 + 1) \% 17 = 6$$

$$215 \% 17 = 11 \quad \text{hit} \Rightarrow (11 + 1 + 1) \% 17 = 13$$

$$30 \% 17 = 13 \quad \text{hit} \Rightarrow (13 + 1 + 1) \% 17 = 15$$

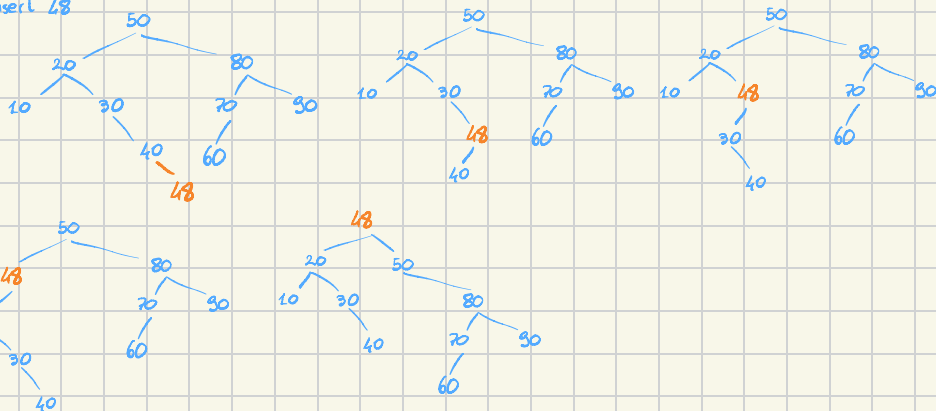
$$36 \% 17 = 11 \quad \text{hit} \Rightarrow (11 + 1 + 1) \% 17 = 13 \quad \text{hit} \rightarrow (11 + 2 + 4) \% 17 = 17 \% 17 = 0$$

Es. 5) Inserire in radice BST 48 e 78, poi cancellare 48.

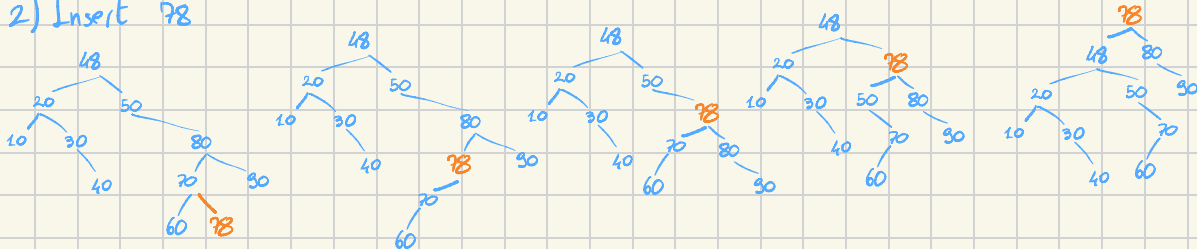


SVOLGIMENTO:

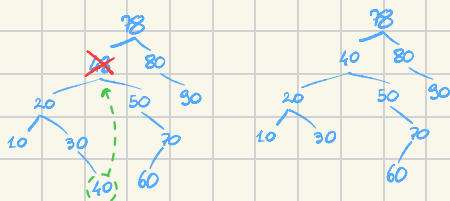
1) Insert 48



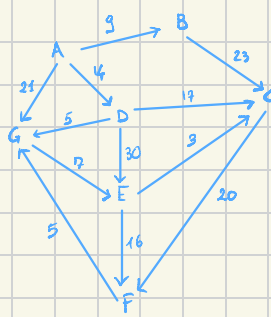
2) Insert 78



3) Delete 48 (predecessore)



Es. 6) Dij Kstra da A



SVOLGIMENTO:

- P₀: PQ: { A/0, B/∞, C/∞, D/∞, E/∞, F/∞, G/∞ }
 P₁: PQ: { B/9, D/14, G/21, C/∞, E/∞, F/∞ } A = 0
 P₂: PQ: { D/14, G/21, C/32, E/∞, F/∞ } B = 9
 P₃: PQ: { G/19, C/31, E/44, F/∞ } D = 14
 P₄: PQ: { E/26, C/31, F/∞ } G = 19
 P₅: PQ: { C/29, E/42 } E = 26
 P₆: PQ: { E/42 } C = 29
 P₇: PQ: { } E = 42